Page No.
Date

Exercise - 1.5

Q1.) Find:

a.) $(64)^{\frac{1}{2}}$ ⟶ Denominator → write the base with power equal to denominator & then apply $(a^p)^q = (a)^{pq}$

$\Rightarrow (8^2)^{\frac{1}{2}}$

$= 8^{\cancel{2} \times \frac{1}{\cancel{2}}}$

$= 8.$

b.) $32^{\frac{1}{5}}$

$\Rightarrow (2^5)^{\frac{1}{5}}$

$\Rightarrow (2)^{\cancel{5} \times \frac{1}{\cancel{5}}}$

$\Rightarrow 2$

c.) $(125)^{\frac{1}{3}}$

$\Rightarrow (5^3)^{\frac{1}{3}}$

$\Rightarrow (5)^{\cancel{3} \times \frac{1}{\cancel{3}}}$

$\Rightarrow 5$

Q2.) Find:

a. $(9)^{\frac{3}{2}}$

$= (3)^{\cancel{2} \times \frac{3}{\cancel{2}}}$

$= 3^3$

$= 27$

b. $(32)^{\frac{2}{5}}$

$= (2)^{\cancel{5} \times \frac{2}{\cancel{5}}}$

$= 2^2$

$= 4$

c. $(16)^{\frac{3}{4}}$

$= (2)^{\cancel{4} \times \frac{3}{\cancel{4}}}$

$= 2^3$

$= 8$

d. $(125)^{-\frac{1}{3}}$

$= (5)^{\cancel{3} \times -\frac{1}{\cancel{3}}}$

$= (5)^{-1}$

$= \frac{1}{5}$

Q3. simplify

a. $2^{\frac{2}{3}} \cdot 2^{\frac{1}{5}}$

$= (2)^{\frac{2}{3} + \frac{1}{5}}$

$= (2)^{\frac{10+3}{15}}$

$= (2)^{\frac{13}{15}}$

b. $\left(\frac{1}{33}\right)^7$

$= \frac{1}{3^{21}}$

c. $\frac{11^{\frac{1}{2}}}{11^{\frac{1}{4}}}$

$= (11)^{\frac{1}{2} - \frac{1}{4}}$

$= (11)^{\frac{2-1}{4}}$

$= (11)^{\frac{1}{4}}$

d. $7^{\frac{1}{2}} \cdot 8^{\frac{1}{2}}$

$= (7 \times 8)^{\frac{1}{2}}$

$= (56)^{\frac{1}{2}}$

Date

Q14. $\sqrt[4]{81^{-2}}$ $\left[\frac{1}{81}^{-2} \quad \left(\frac{1}{9}\right)^{4 \times \frac{1}{4}} = \frac{1}{9}\right]$

sol: $(81^{-2})^{\frac{1}{4}}$

$\Rightarrow (81)^{-\cancel{2} \times \frac{1}{\cancel{4}_2}}$

$\Rightarrow (81)^{-\frac{1}{2}}$

$\Rightarrow (9)^{\cancel{2} \times -\frac{1}{\cancel{2}}}$

$\Rightarrow (9)^{-1}$

$\Rightarrow \frac{1}{9}$

Q164.

a.) $\left[5\left(8^{\frac{1}{3}} + 27^{\frac{1}{3}}\right)^3\right]^{\frac{1}{4}}$

$\Rightarrow \left[5\left((2)^{\cancel{3} \times \frac{1}{\cancel{3}}} + (3)^{\cancel{3} \times \frac{1}{\cancel{3}}}\right)^3\right]^{\frac{1}{4}}$

$\Rightarrow \left[5(2+3)^3\right]^{\frac{1}{4}}$

$\Rightarrow \left[5\ (5)^3\right]^{\frac{1}{4}}$

$\Rightarrow \left[5 \times 125\right]^{\frac{1}{4}}$

$\Rightarrow \left[225\right]^{\frac{1}{4}}$

$\Rightarrow [(\cancel{125}\ 45)^{\frac{1}{\cancel{4}}} = 5$

164.

4.) $\left[\left((625)^{-\frac{1}{2}}\right)^{-\frac{1}{4}}\right]^2$

$\Rightarrow \left[\left((25)^{\cancel{2} \times -\frac{1}{\cancel{2}}}\right)^{-\frac{1}{4}}\right]^2$

$\Rightarrow \left[\left((25)^{-1}\right)^{-\frac{1}{4}}\right]^2$

Spiral

Page No.

Date

$\left[\left(\frac{1}{25}\right)^{-\frac{1}{4}}\right]^2$

$\left[(25)^{\frac{1}{4}}\right]^2$

$\Rightarrow \left((\cancel{25}5)^{\cancel{2} \times \frac{1}{\cancel{4}_2}}\right]^2$

$\Rightarrow (5)^{\frac{1}{\cancel{2}} \times \cancel{2}}$

$\Rightarrow 5$

Q. $64^{-\frac{1}{3}}\left[64^{\frac{1}{3}} - 64^{\frac{2}{3}}\right]$

$\Rightarrow (4)^{\cancel{3} \times -\frac{1}{\cancel{3}}}\left[\cancel{(64)^{\frac{1}{3}-\frac{2}{3}}}\right]\left[4 - 16\right]$

$\Rightarrow \frac{1}{\cancel{4}} \times -\cancel{12} = -3$

Q. $\frac{9^{\frac{1}{3}} \times 27^{-\frac{1}{3}}}{3^{\frac{1}{6}} \times 3^{-\frac{4}{3}}}$

sol: $\frac{9^{\frac{1}{3}} \times 27\ 3^{-\frac{4}{3}}}{3^{\frac{1}{6}} \times 27^{\frac{1}{3}}}$

$\Rightarrow \frac{(3)^{\frac{2}{3}} \times (3)^{\frac{4}{3}}}{(3)^{\frac{1}{6}} \times (3)^1} = \frac{(3)^{\frac{2}{3}+\frac{4}{3}}}{(3)^{\frac{1}{6}+1}} = \frac{(3)^{\frac{\cancel{6}2}{\cancel{3}}}}{(3)^{\frac{1+6}{6}}}$

$= \frac{(3)^2}{(3)^{\frac{7}{2}}}$ $\Rightarrow (3)^{2-\frac{7}{2}}$

$\Rightarrow (3)^{\frac{4-7}{2}}$

$\Rightarrow (3)^{-\frac{3}{2}}$

Spiral

Page No.
Date

Q.) $\dfrac{9^{\frac{1}{3}} \times 27^{-\frac{1}{2}}}{3^{\frac{1}{6}} \times 3^{-\frac{2}{3}}}$

sol:

$$\frac{9^{\frac{1}{3}} \times 27^{-\frac{1}{2}}}{3^{\frac{1}{6}} \times 3^{-\frac{2}{3}}} = \frac{(3)^{\frac{2}{3}} \times (3)^{-\frac{3}{2}}}{(3)^{\frac{1}{6}} \times (3)^{-\frac{2}{3}}}$$

$$\Rightarrow \frac{(3)^{\frac{2}{3} + \left(-\frac{3}{2}\right)}}{(3)^{\frac{1}{6} + \left(-\frac{2}{3}\right)}} \Rightarrow \frac{(3)^{\frac{4-9}{6}}}{(3)^{\frac{1-4}{6}}}$$

$$\Rightarrow \frac{(3)^{-\frac{5}{6}}}{(3)^{-\frac{3}{6}}} \Rightarrow (3)^{\frac{5}{6} - \left(-\frac{3}{6}\right)} \Rightarrow (3)^{-\frac{5}{6} + \frac{3}{6}}$$

$$\Rightarrow (3)^{-\frac{2}{6}} = \frac{1}{3^{\frac{1}{3}}} \quad \text{Ans}$$

Q. $\left(\frac{8}{5}\right)^{-12} \times \left(\frac{3}{5}\right)^{4} \times \left(\frac{32}{5}\right)^{6}$

sol: $\left(\frac{8}{81}\right)\left(\frac{8}{8}\right)\left(\frac{8}{8}\right)\left(\frac{8}{8}\right)\left(\frac{8}{8}\right)\left(\frac{8}{8}\right)\left(\frac{8}{8_2}\right)\left(\frac{5}{8_2}\right)\left(\frac{5}{8_2}\right)\left(\frac{5}{8_2}\right)\left(\frac{5}{8_2}\right)\left(\frac{5}{8_2}\right) \times$

$\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right) \times \left(\frac{32}{5}\right)\left(\frac{32}{5}\right)\left(\frac{32}{5}\right)\left(\frac{32}{5}\right)\left(\frac{32}{5}\right)\left(\frac{32}{5}\right)$

$$\Rightarrow \frac{(3\times3\times3\times3) \times (5\times5)}{2\times2\times2\times2\times2\times2} = \frac{81 \times 25}{64} = \frac{2025}{64}$$

$$\frac{8^{-12}}{5^{-12}} \times \frac{3^{4}}{5^{4}} \times \frac{32^{6}}{5^{-6}}$$

Page No.
Date

Questions

Q.1. Find the value of $(1^3 + 2^3 + 3^3 + 8^2)^{-5/2}$

$\Rightarrow (1 + 8 + 27 + 64)^{-5/2}$

$\Rightarrow (9 + 91)^{-5/2}$

$\Rightarrow (100)^{-5/2}$

$\Rightarrow (10)^{\cancel{2} \times -5/\cancel{2}}$

$\Rightarrow \frac{1}{10^5} = \frac{1}{100000}$

Q2. Find the value of x, if $x^{1/2} = (36)^{0.5}$

sol: $x^{\frac{1}{2}} = (36)^{\frac{0.5}{1/2}}$

$x^{1/2} = (36)^{\frac{1}{2}}$, $x = 36$

~~$x^{1/2} = (6)^{\cancel{2} \times \frac{1}{\cancel{2}}}$~~

~~$\boxed{x = 36}$~~

Q3. Find the value of x, if $(\sqrt{3})^x = 3^7$

sol: $(3^{\frac{1}{2}})^x = 3^7$

$(3^{x/2}) = 3^7$ (base same, power diff.)

$\frac{x}{2} = 7$

$x = 7 \times 2 = \underline{\underline{14}}$

Q4. if $2^{5x} \div 2^x = \sqrt[5]{32}$, then find x

sol: $(2)^{5x - x} = \sqrt[5]{32}$ $(a^m \div a^n = a^{m-n})$

$\Rightarrow (2)^{4x} = (32)^{\frac{1}{5}}$

$\Rightarrow (2)^{4x} = (2)^{\cancel{5} \times \frac{1}{\cancel{5}}}$

$\Rightarrow (2)^{4x} = 2^1$

$4x = 1$

$x = \frac{1}{4}$

$\Rightarrow \boxed{m = n}$ when, m is even $\boxed{a = \pm b}$

Date

Q5. Evalute

- $a^{x-y} \cdot a^{y-z} \cdot a^{z-x}$

sol: $(a)^{x-y+y-z+z-x}$

$\Rightarrow a^0 = 1$

Q.6 simplify: $(12)^{2/5} \cdot (5)^{2/5}$ $\quad (a^m \times b^m = (a \times b)^m)$

sol:

$(12 \times 5)^{2/5}$

$(60)^{2/5}$

Q.7 Evalute:

1.) $\dfrac{2^{38} + 2^{37} + 2^{36}}{2^{39} + 2^{36} + 2^{37}}$

Ans $\dfrac{2^{36}\cancel{(2^2 + 2 + 1)}}{2^{37}\cancel{(2^2 + 2 + 1)}} = 2^{36-37} = 2^{-1} = \dfrac{1}{2}$

Highest common lena h

2.) $\left[(64)^{1/6}\right]^2$

sol: $\left[(8)^{1/6}\right]^2$

$\Rightarrow (8)^{\frac{2}{6}}$

$\Rightarrow (8)^{\frac{1}{3}}$

$\Rightarrow (2)^{3 \times \frac{1}{3}}$

$\Rightarrow 2$

Q8. simplify

$$\frac{(25)^{3/2} \times (243)^{3/5}}{(16)^{5/4} \times (8)^{4/3}}$$

sol: $\dfrac{(5)^{2 \times \frac{3}{2}} \times (3)^{5 \times \frac{3}{5}}}{(2)^{4 \times \frac{5}{4}} \times (2)^{3 \times \frac{4}{3}}}$

$\Rightarrow \dfrac{125 \times 27}{32 \times 16} = \dfrac{3375}{512}$

Q9.) $5^{2x-1} - (25)^{x-1} = 2500$, find x.

sol: $5^{2x-1} - (5)^{2x-2} = 2500$

$\cancel{(2x-1)}$

$\cdot 5^{2x-2}(5-1) = 2500 \Rightarrow 5^{2x-2}(4) = 2500 \Rightarrow 5^{2x-2} \times 4$

$= 2500$ 625

(Imp.)

$\Rightarrow 5^{2x-2} = 5^4$

$2x - 2 = 4$

$2x = 6$

$\boxed{x = 3}$

Q10. Find 'x'

1.) $25^{2x-3} = 5^{2x+3}$

$\Rightarrow (5)^{4x-6} = (5)^{2x+3}$

$\Rightarrow 4x-6 = 2x-3$

$\Rightarrow 4x-2x = +3+6$

$\Rightarrow 2x = 9$

$x = \frac{9}{2}$

2.) $(4)^{2x-1} - (16)^{x-1} = 384$

$\Rightarrow (4)^{2x-1} - (4)^{2x-2} = 384$

$\Rightarrow 4^{2x-2}(4-1) = 384$

$\Rightarrow 4^{2x-2}(3) = \cancel{384}\ 128$

$\Rightarrow 4^{2x-2} = 128$

$\Rightarrow 2^{4x-4} = 2^7$

$4x-4 = 7$

$4x = 7+4$

$4x = 11$

$x = \frac{11}{4}$

$$\left[\begin{array}{l} \frac{4^{2x-1}}{4^{2x-2}} \\ 2x-1-(2x-2) \\ 2x-1-2x+2 \end{array}\right]$$

Q11. Evalute -

$$\frac{(4^{9/6})}{4^a} \times \frac{2^{2a+1}}{2^{a-1}}$$

Q12. simplify : $\left(\frac{x^a}{x^{-b}}\right)^{a-b} \cdot \left(\frac{x^b}{x^{-c}}\right)^{b-c} \cdot \left(\frac{x^c}{x^{-a}}\right)^{c-a}$

Q13. $(256)^{0.4} \cdot (256)^{0.6}$

Answers

11.) $\frac{64^{9/6}}{4^a} \times \frac{2^{2a+1}}{2^{a-1}}$

Sol: $\frac{(2)^{6\times\frac{9}{6}}}{(2)^{2a}} \times \frac{2^{2a+1}}{2^{a-1}}$

$\Rightarrow \frac{2^9}{2^{2a}} \times \frac{2^{2a+1}}{2^{a-1}}$ $\quad \left[\begin{array}{l} a^m \times a^n = a^{m+n} \\ a^m \times a^n = a^{m+n} \end{array}\right]$

$\Rightarrow \frac{(2)^{9+2a+1}}{(2)^{2a+(a-1)}}$

$\Rightarrow \frac{(2)^{3a+1}}{(2)^{3a-1}} \Rightarrow (2)^{3a+1-(3a-1)}$ $\quad [a^m/a^n = a^{m-n}]$

$\Rightarrow (2)^{\cancel{3a}+1-\cancel{3a}+1}$

$\Rightarrow (2)^2 = 4$ Ans

2nd

$\frac{(2)^9}{(2)^{2a}} \times \frac{(2)^{2a+1}}{(2)^{a-1}}$

$\Rightarrow (2)^{-a} \times (2)^{a+2}$

$\Rightarrow (2)^{\cancel{-a}+\cancel{a}+2}$

$\Rightarrow (2)^2 = 4$